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PROFESSIONAL SUMMARY

- Data Engineer with **3 years of hands-on experience** in developing and supporting scalable data processing and ETL pipelines using **Python, PySpark, Apache Spark, and AWS**.
- Experienced in developing **PySpark-based ETL pipelines on AWS EMR** to ingest, transform, cleanse, validate, and load large volumes of structured and semi-structured data.
- Hands-on experience working with **Amazon S3** for data ingestion and storage, including organizing datasets into **raw, cleansed, and curated data layers**.
- Proficient in processing multiple data formats including **CSV, JSON, Parquet, and ORC**, with experience defining explicit schemas and handling nested and semi-structured data.
- Skilled in implementing PySpark transformations including **joins, aggregations, filtering, deduplication, window functions, data cleansing, schema alignment, and conditional transformations**.
- Experienced in developing **full-load and incremental-load ETL workflows**, including watermark-based processing, change detection, Slowly Changing Dimension (SCD) logic, and handling late-arriving data.
- Hands-on experience implementing **data quality and validation checks**, including duplicate detection, mandatory-field validation, referential integrity, source-to-target reconciliation, and rejected-record handling.
- Experienced in **Spark performance optimization**, including broadcast joins, repartitioning, coalesce, caching, partition tuning, reducing unnecessary data processing, and addressing data-skew issues.
- Familiar with analyzing **Spark execution plans and Spark UI** to identify expensive joins, shuffles, resource bottlenecks, and performance issues.
- Experienced in configuring and managing **AWS EMR Spark applications**, including EMR Steps, cluster configuration, executor/driver resource tuning, bootstrap actions, and job troubleshooting.
- Hands-on experience orchestrating data workflows using **Apache Airflow**, including EMR cluster creation, Spark job submission, step monitoring, retries, dependencies, and failure handling.
- Experienced in integrating **Airflow with AWS S3, EMR, Lambda, and SNS** for automated and event-driven ETL workflows.
- Familiar with **AWS IAM roles and S3 access policies** for providing secure access to data processing workloads.
- Experienced in troubleshooting production ETL issues using **Spark logs, EMR/YARN logs, AWS CloudWatch, and application error logs**, along with supporting controlled backfills and data recovery activities.
- Strong understanding of **ETL concepts, distributed data processing, data partitioning, fault tolerance, workflow orchestration, and production data pipeline support**.

- **ETL pipelines in Databricks**, developing DataFrame and Spark SQL transformations to ingest, cleanse, deduplicate, enrich, and process structured and semi-structured data stored in **AWS S3**.
- Worked on **Delta Lake-based incremental and batch processing**, including MERGE/upsert operations, data validation, partitioning, and Spark performance tuning using techniques such as **broadcast joins, repartition/coalesce, caching, and Spark UI analysis**.

TECHNICAL SKILLS

Primary Skills : Amazon EMR, PySpark, Spark SQL, Amazon S3

Cloud Platforms : AWS, Databricks

Data Processing : Apache Spark, Distributed Data Processing, Spark SQL

EMR Components : EMR Cluster Provisioning, EMR Steps, YARN Resource Management, Auto Scaling Policies

Workflow Management : Apache Airflow, AWS Step Functions, EMR Job Scheduling

Data Storage : Amazon S3, Parquet, JSON, CSV

Database Technologies : MS SQL Server, MySQL, Amazon RDS

Languages : SQL, Pyspark

Monitoring & Optimization : Spark Optimization, Data Validation, Job Monitoring, Error Handling, Production Support

Operating Environments : Linux, CentOS, Windows

PROFESSIONAL EXPERIENCE

PROJECT

Project Name: Capital One Sep 2023– Till Date

Project Role: Data Engineer

Responsibilities:

- Developed and maintained batch ETL pipelines using **PySpark DataFrames, Spark SQL, and AWS EMR** to transform raw data into analytics-ready datasets.
- Built data ingestion workflows to process **structured and semi-structured data** from Amazon S3 and relational databases using JDBC.
- Implemented data processing across **raw, cleansed, and curated layers** in Amazon S3 using CSV, JSON, Parquet, and ORC file formats.
- Applied PySpark transformations including **data cleansing, standardization, deduplication, filtering, joins, aggregations, and window functions** based on business requirements.
- Defined explicit schemas and handled **nested JSON structures, null values, invalid records, inconsistent formats, and data type standardization** during ingestion.
- Developed reusable and parameterized PySpark components to support **different processing dates, source paths, environments, and target locations**.
- Implemented **full-load and incremental-load processing**, including watermark-based processing and change detection using business keys and attribute comparisons.

- Supported **Slowly Changing Dimension (SCD)** processing and handled late-arriving records through controlled partition-level reprocessing.
- Implemented data quality and reconciliation checks using **record counts, duplicate validation, mandatory-field checks, aggregate comparisons, and key-level validation**.
- Separated rejected records from valid datasets and stored validation results in **Amazon S3** for investigation and operational tracking.
- Optimized PySpark jobs using **broadcast joins, repartition/coalesce, caching, early filtering, partition tuning, and built-in Spark functions** to improve processing efficiency.
- Analyzed **Spark UI, execution plans, driver/executor logs, and YARN logs** to identify performance bottlenecks, expensive joins, shuffles, and job failures.
- Configured and submitted PySpark applications through **AWS EMR Steps and spark-submit**, including appropriate cluster and Spark resource configurations.
- Worked with **EMR bootstrap actions, executor/driver resource settings, and cluster configurations** based on workload requirements.
- Orchestrated EMR-based ETL workflows using **Apache Airflow**, including job dependencies, retries, monitoring, and failure handling.
- Integrated curated S3 datasets with the **AWS Glue Data Catalog and Amazon Athena** for downstream data discovery and querying.
- Worked with **IAM roles and S3 access policies** to support secure access to data and AWS resources required by ETL jobs.
- Supported production data pipelines by performing **job monitoring, error analysis, controlled backfills, data validation, and incident resolution**.
- Maintained ETL documentation covering **transformation logic, data validation rules, processing workflows, and operational procedures**.

WORK EXPERIENCE

A&N SOFTWARE SOLUTIONS– Sep 2023– Till Date

EDUCATION

Institute/College	Duration	Percentage Obtained
Vishwa Vishwani School of Business	2021-23	72%
Sree Nidhi Degree College	2018-21	85%
Narayana Junior College	2016-18	89%
ST Frances EM School	2015-16	75%